Methane is invisible. Its consequences are not.

46 miners died on **28 October 2023** when a fire, presumably caused by a methane explosion, broke out in the **Kostenko coal mine**, Karaganda region, Kazakhstan.

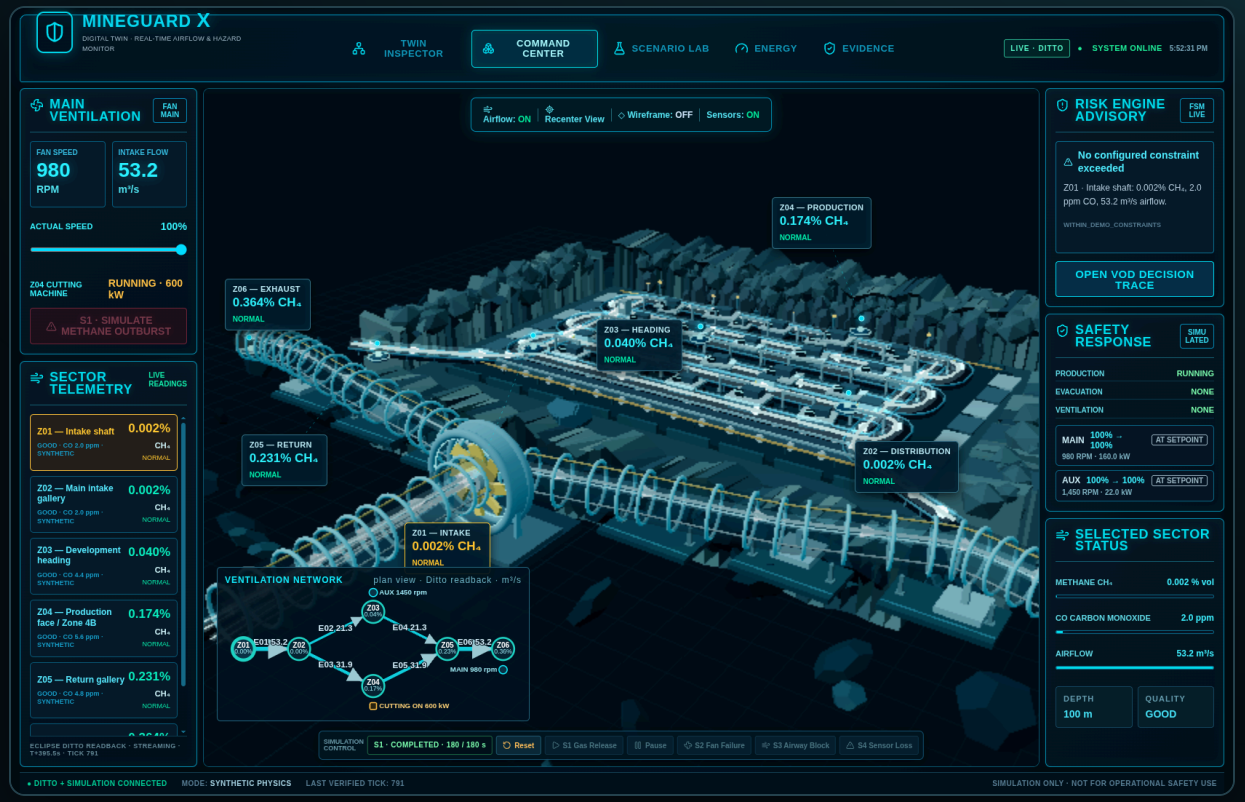
In a gassy mine, ventilation keeps methane diluted — and every fan change moves air in every connected roadway.

MineGuard X

A physics-informed digital twin of an underground coal-mine ventilation section, built on OpenEgiz and Eclipse Ditto.

SIMULATION ONLY

FICTIONAL MINE · DOCUMENTED PARAMETERS



16 Ditto twins

4 what-if scenarios

3 users, 3 decisions

252 automated tests

Accident figures: “Kostenko mine disaster”, Wikipedia (final toll 46; early reports on 28–29 Oct 2023 gave lower counts while recovery continued)

CONCEPT

Three users, three questions — one coupled system

Control-room operator

“What is happening now, and where?”

Live CH_4, CO and airflow per zone, the risk state with its reasons, and whether the simulated production stop or evacuation advisory is active.

Ventilation engineer

“Which fan setting is safe, and how much energy does it use?”

Every candidate fan setting is forecast 60 s ahead; unsafe ones are rejected and the lowest-energy safe one is proposed.

Reliability engineer

“What happens when something fails — and did the response actually happen?”

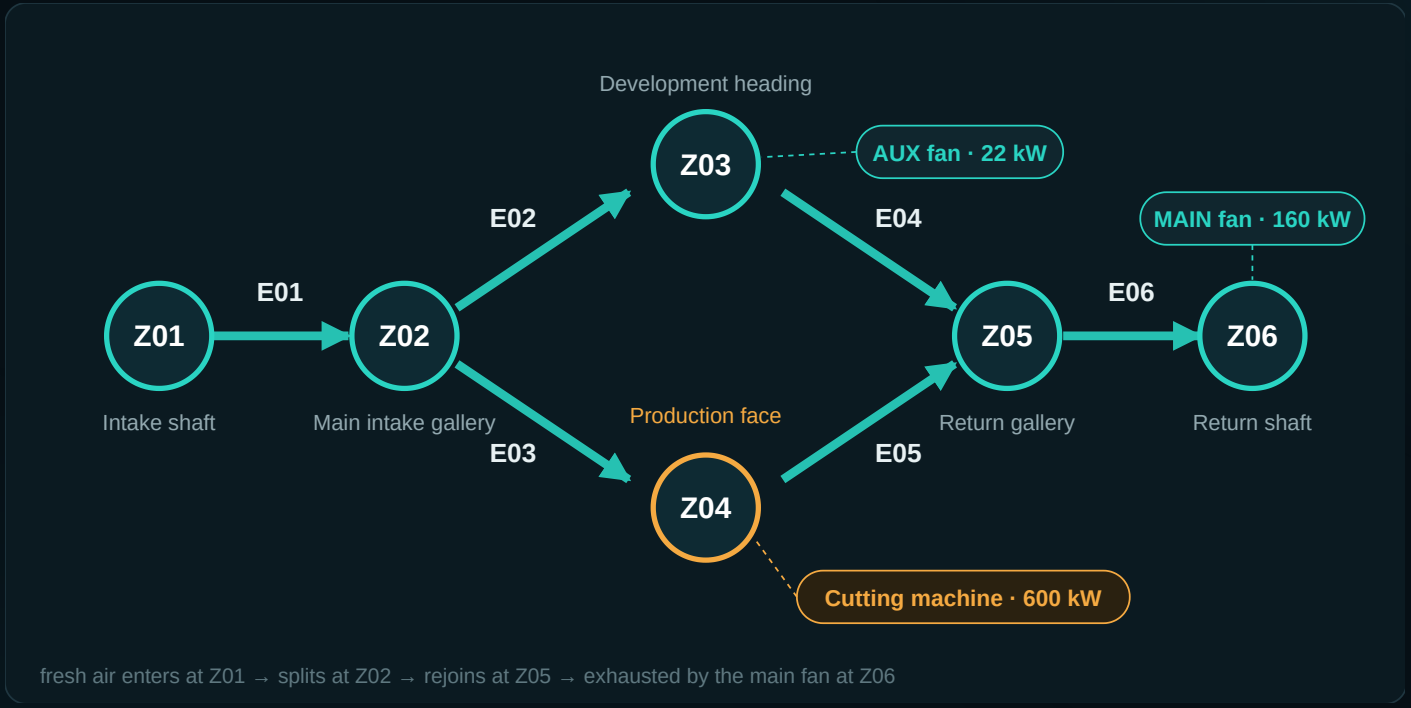
Fan failure, roof fall and sensor loss scenarios, and requested → actual feedback for every fan command.

Fan speed changes → **Airflow** in every connected roadway → **Methane dilution** in every zone → **Risk state** and simulated safety actions → **Energy** fan power ∝ speed³

Gas risk, fan reliability and electricity use cannot be answered sensor by sensor — a connected twin can. Main fans are large electrical loads, but **safety always outranks energy**.

THE MINE

One gassy coal-mine section, modeled as 16 Eclipse Ditto twins



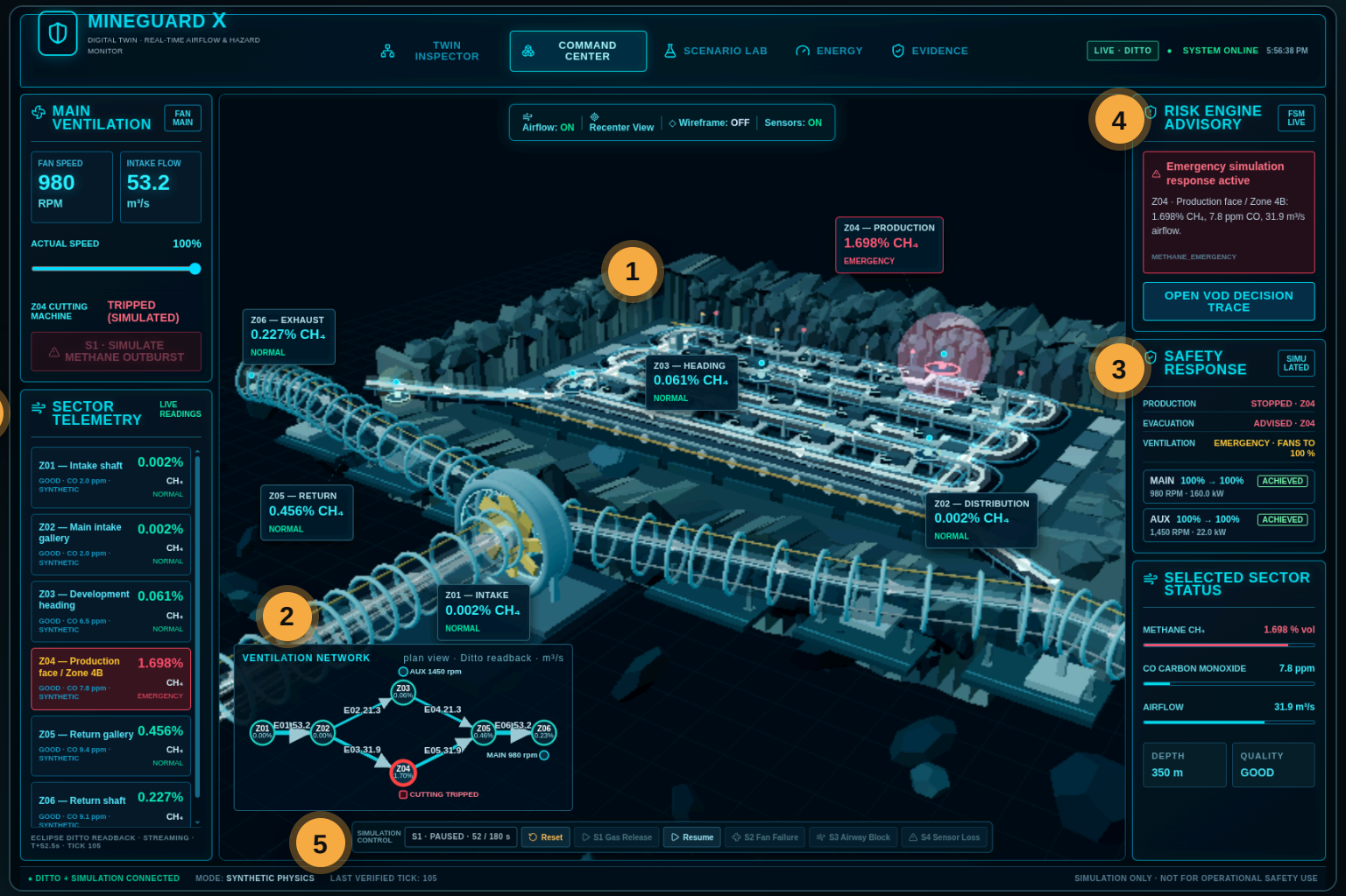
fresh air enters at Z01 → splits at Z02 → rejoins at Z05 → exhausted by the main fan at Z06

TWIN TYPE	COUNT	LIVE STATE
`mine-01`	1	max CH_4, max CO, total airflow, fan power, tick
`zone-z01…z06`	6	CH_4 (% vol), CO (ppm), airflow, risk, simulated interlock
`airway-e01…e06`	6	flow (m^3/s), obstruction
`fan-main` · `fan-aux`	2	speed, requested speed, rpm, kW, RUNNING / FAULTED
`shearer-z04`	1	RUNNING / TRIPPED, kW, cutting CH_4 and CO

Relations are Ditto attributes: each airway names its from- and to-zone; each zone lists its airways, neighbours, fans and equipment; all twins carry `_parents: mineguard:mine-01`. Inspect them live at `/twins`.

COMMAND CENTER · HTTP://127.0.0.1:8091

Everything on screen is read back from Eclipse Ditto



1. **3D cutaway**
   risk colours per zone, a methane volume that grows with CH_4, airflow particles, fan rotor at actual RPM

2. **Ventilation network**
   airways E01–E06 with live m³/s; a blocked airway turns dashed amber with its multiplier

3. **Safety Response**
   production, evacuation, ventilation; each fan's requested → actual % with PENDING / ACHIEVED / FAULTED

4. **Risk engine advisory**
   the worst zone, its readings and reason codes

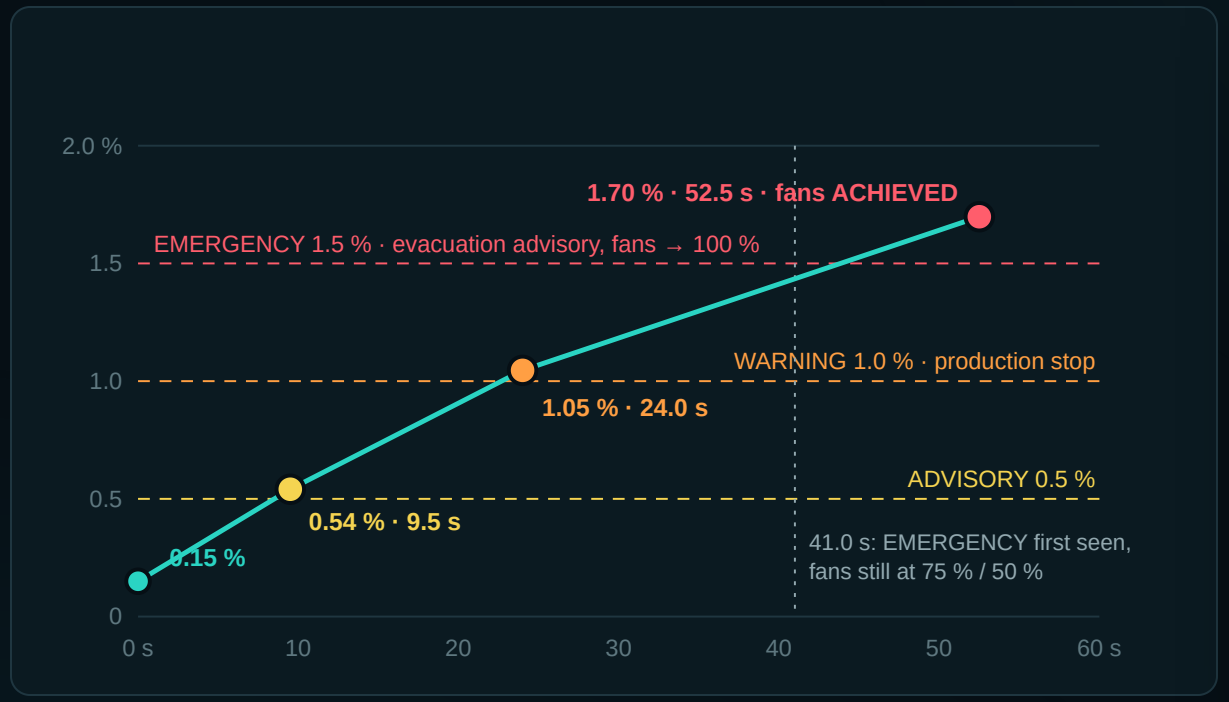
5. **Scenario tray**
   S1–S4, the active run and its time, Pause and Reset

6. **Zone list**
   CH_4, CO and risk for all six zones

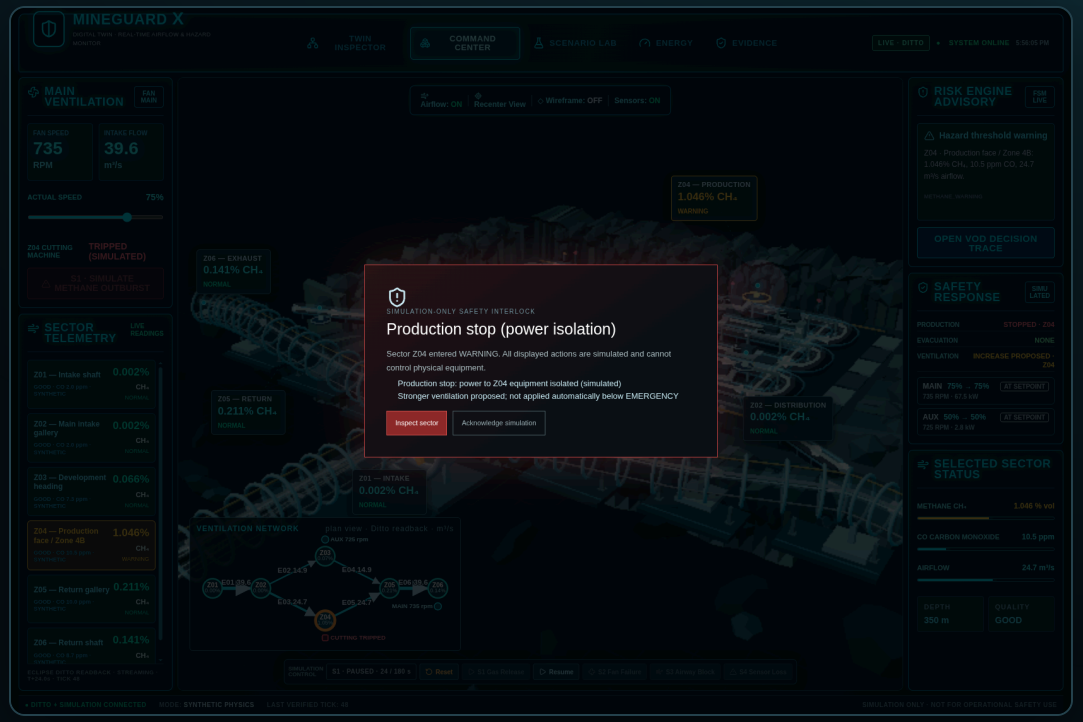
Screenshot: S1 at EMERGENCY with both fans ACHIEVED · software WebGL (SwiftShader) with the lite render profile

WHAT-IF S1 · PRODUCTION-FACE METHANE RELEASE

Methane crosses every threshold — and the response follows



Z04 CH_4 sampled from the 1 Hz Ditto readback (straight lines join the samples; 0.15 % is the initial state)

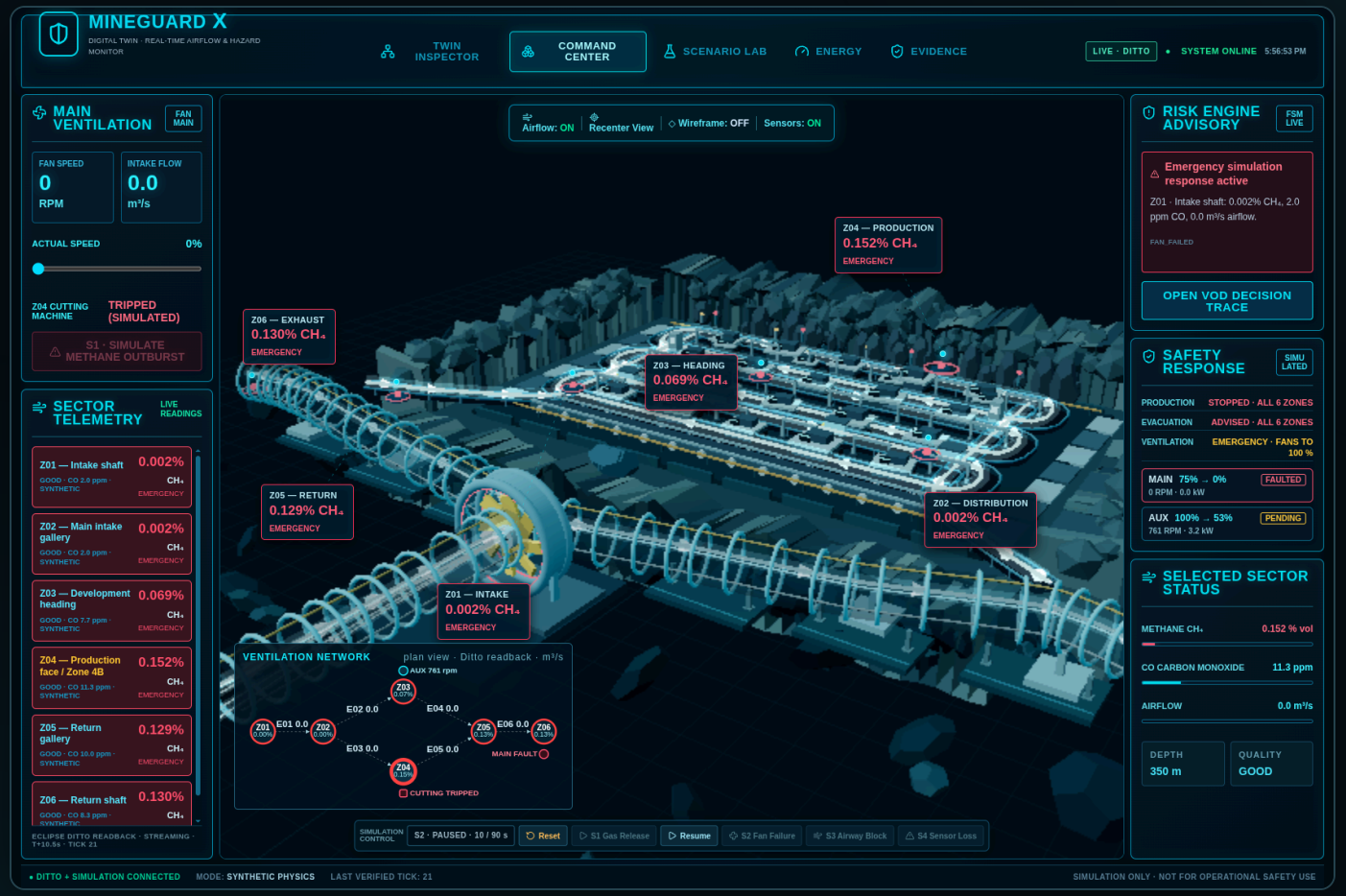


WARNING: the popup names only the production stop

The Z04 source rises to 0.8 m³/s for 180 s; methane accumulates through the mass balance. WARNING trips the cutting machine (its cutting methane leaves the balance). EMERGENCY adds the evacuation advisory and drives both fans to 100 % at 5 %/s until Ditto reads them back **ACHIEVED**.

Undo: Reset restores the exact tick-0 baseline.

Losing the main fan puts every zone into EMERGENCY



Ditto twin mineguard:fan-main read back FAULTED by the scenario evidence run

0 RPM — At T+10 s the main fan faults; Ditto reads back **FAULTED**.

0 m³/s — Network airflow stops in all six airways.

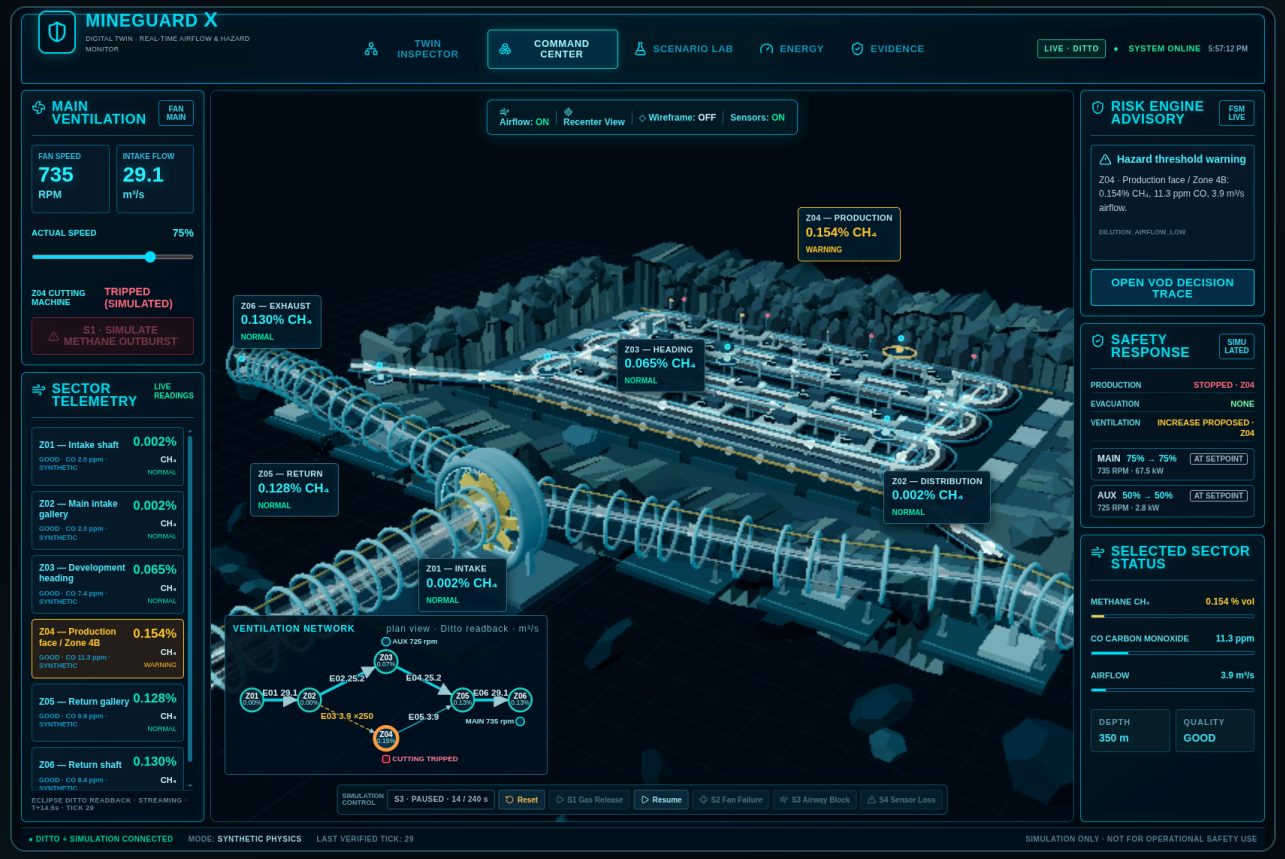
6 / 6 — Zones in EMERGENCY with reason `FAN_FAILED`; the cutting machine trips.

AUX 100 % — The failed fan is excluded (`FAILED_FAN_EXCLUDED`); the healthy fan is commanded up.

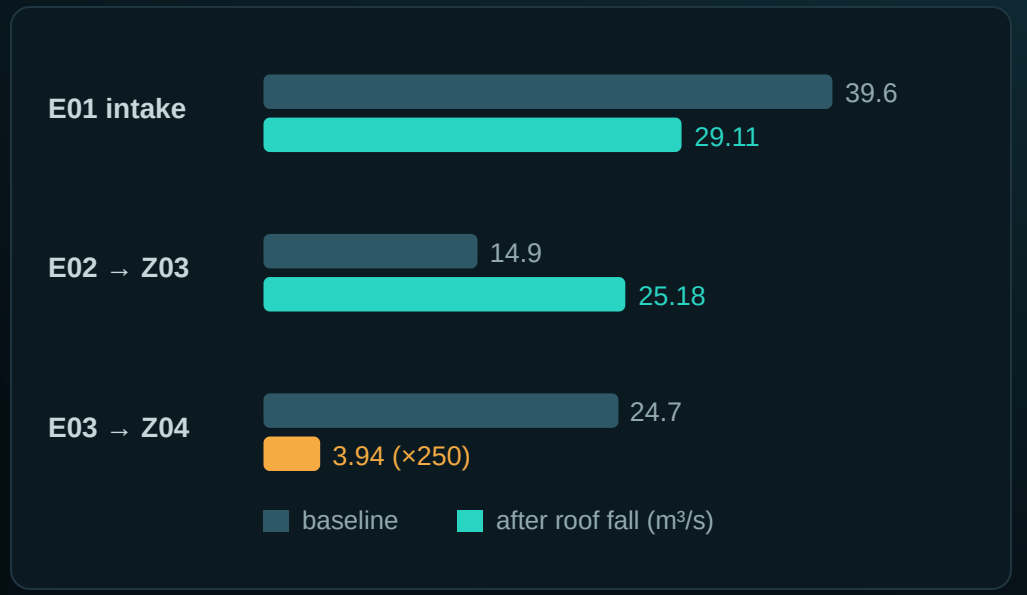
VOD off — The optimizer is disabled: safety outranks energy.

Undo: Reset, or the fault clears at T+90 s; the tray then shows `COMPLETED` and keeps Reset.

The network redistributes air — and the face loses its dilution flow



The network map draws E03 dashed amber with "×250"



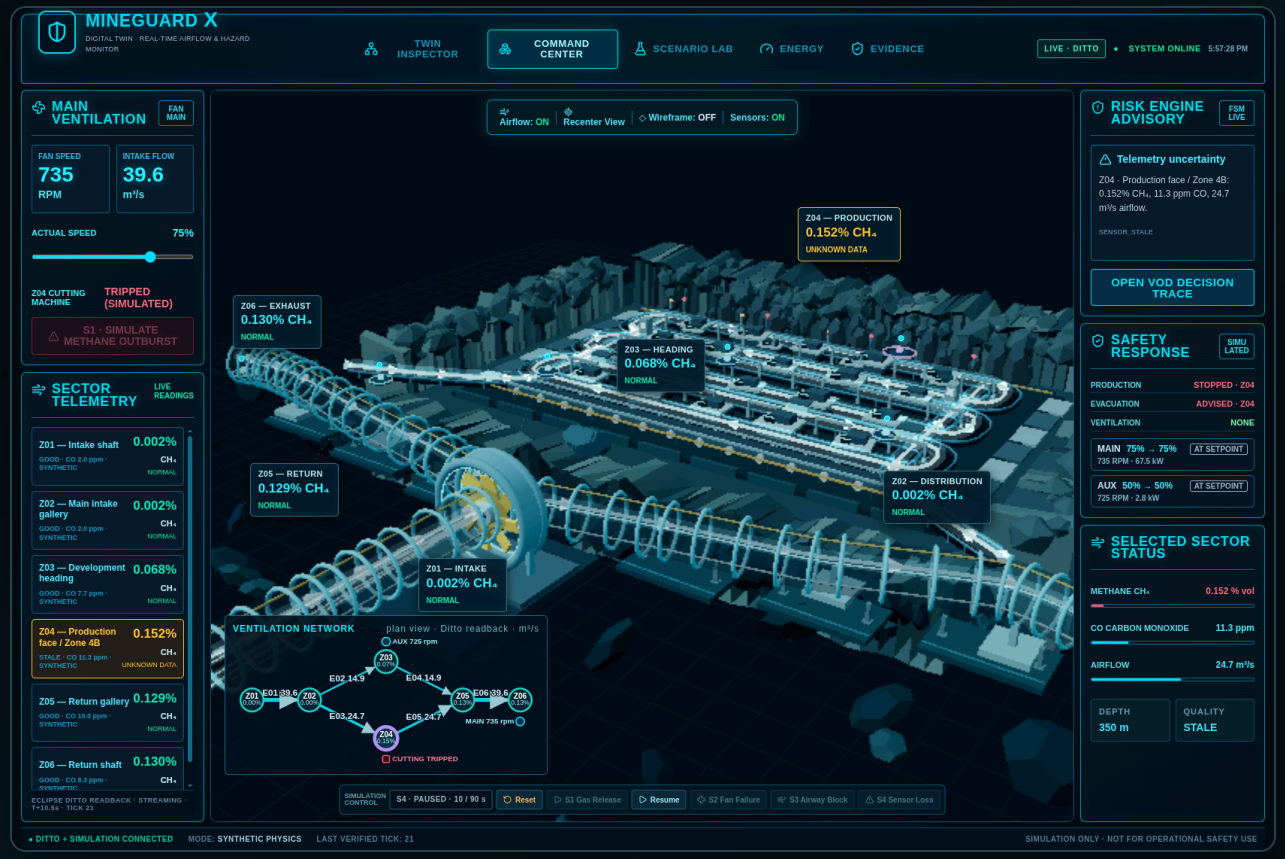
At T+10 s the resistance of E03 is multiplied by 250. Air shifts to the heading branch and total intake falls. The face drops below the **5 m³/s** dilution constraint → WARNING (`DILUTION_AIRFLOW_LOW`), the cutting machine trips, and methane rises.

Undo: Reset, or the obstruction clears at 240 s.

Baseline flows: SUBMISSION.md · roof-fall flows: report.json of the scenario evidence run (Ditto readback)

WHAT-IF S4 · PRODUCTION-FACE GAS SENSOR LOSS

A silent sensor is never a safe reading



Ditto twin mineguard:zone-z04 read back with quality STALE

T+10 s · Z04 CH_4 and CO sensors stop reporting

↓

last value kept, marked **STALE**

↓

risk becomes **UNKNOWN_DATA** — not NORMAL

↓

simulated **production stop** + **evacuation advisory**, optimizer disabled

Z04 turns purple in the 3D scene and on the network map. The same rule protects the interface: if the readback itself goes stale, the Safety Response panel shows “UNKNOWN — telemetry stale” instead of the last good numbers.

Undo: Reset, or the sensors return at 90 s.

ENERGY · CONSTRAINT-FIRST VENTILATION ON DEMAND

The lowest-energy fan setting that keeps every constraint

check risk and data quality → enumerate ramp-limited fan settings → forecast airflow, CH_4, CO for 60 s → reject every violation → pick minimum energy

Current speeds
main 75 % · aux 50 %

1.171 kWh

Selected VOD
main 60 % · aux 40 %

0.599 kWh

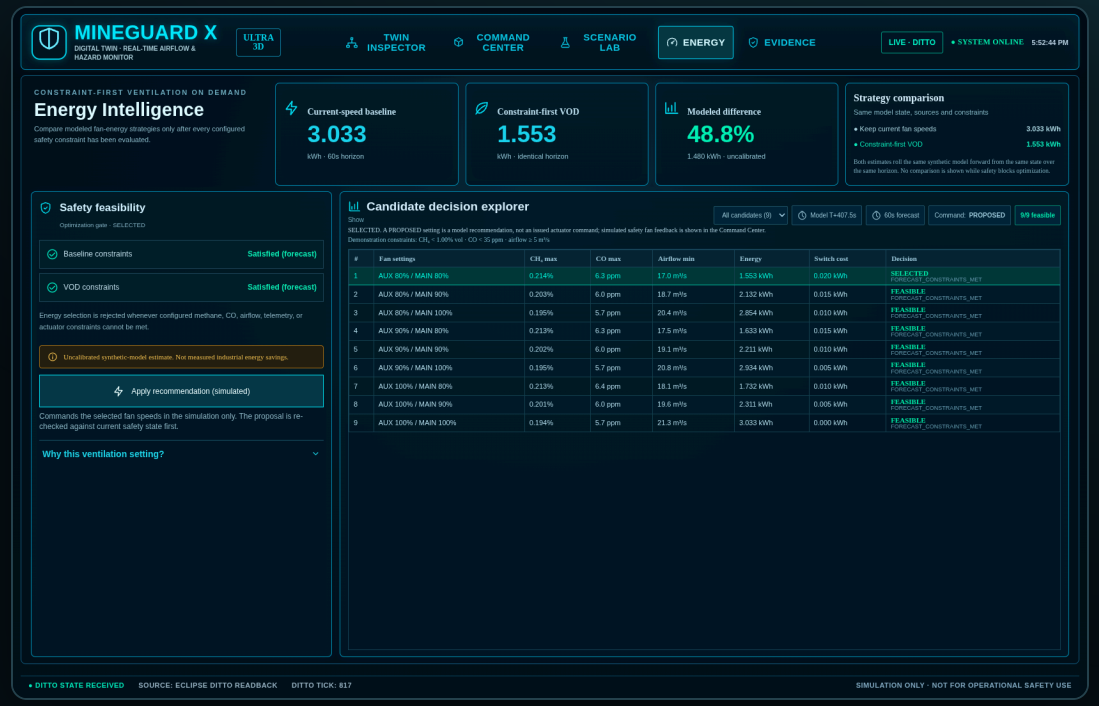
fan energy over the same 60 s horizon, same initial state

48.8 %
model estimate, not measured

16 / 16
candidates feasible

11.9
m³/s min forecast airflow (limit 5)

Synthetic, uncalibrated model estimate of fan energy — not a measured saving.



`/energy captured while both fans ran at 100 %: 3.033 → 1.553 kWh (both → 80 %). Left: the baseline after a reset. Both steps cut each fan by 20 %, so both give 48.8 % (P ∝ s³; 0.8³ = 0.512).`

Selected-setting forecast: max CH_4 0.180 %, max CO 9.9 ppm. **Apply recommendation (simulated)** re-checks safety, then the fans ramp (PENDING until the setpoint). Disabled at WARNING, EMERGENCY and UNKNOWN_DATA.

Real-world context, vendor-reported: ABB reports 54 % less ventilation energy at Boliden Kankberg; an estimated 56 % less fan power is reported for Newmont Éléonore.

HOW IT WORKS

Model → MQTT → Ditto → readback → browser



Built on OpenEgiz

Complete Ditto `modify /features` messages go to the official broker. A publication counts only when all 16 twins return the same run ID and tick; the UI state is rebuilt from that Ditto JSON (`ECLIPSE_DITTO_READBACK`). Scenario commands go to the model, never to equipment.

Physics with explicit units

`airway Δp = R·Q·|Q| · fan Δp = p₀·s² − K·Q² · P = P_rated·s³`

`zone V·dC/dt = G + Σ Q_in·C_in − Σ Q_out·C (CH₄, CO)`

CH_4 is a fraction inside the model and % volume on screen (0.8 % = 8,000 ppm); CO ppm; airflow m^3/s; power kW; energy kWh.

SAFETY RULES

Uncertainty and safety come before optimization

STATE	TRIGGER (DEMONSTRATION)	SIMULATED RESPONSE
NORMAL	all constraints met	optimizer allowed
ADVISORY	CH_4 ≥ 0.5 % or CO ≥ 25 ppm	stronger ventilation proposed (not applied automatically); optimizer still allowed
WARNING	CH_4 ≥ 1.0 %, CO ≥ 35 ppm, or airflow < 5 m^3/s	**production stop**: power to the zone's equipment isolated, cutting machine tripped; optimizer disabled
EMERGENCY	CH_4 ≥ 1.5 %, CO ≥ 50 ppm, or main fan failed	adds **evacuation advisory**; healthy fans commanded to 100 % (5 %/s ramp); failed fan excluded
UNKNOWN_DATA	stale or invalid telemetry	treated as unsafe: production stop and evacuation advisory; never shown as a safe reading

The 1.0 % and 1.5 % methane tiers mirror the action levels of **US 30 CFR 75.323(b)**. The CO tiers reuse 8-hour exposure limits (35 ppm NIOSH REL, 50 ppm OSHA PEL, both time-weighted averages) as instantaneous **demonstration thresholds** — not how those limits are defined, and not a compliance claim.

Downgrades use hysteresis and every transition carries reason codes. Every action is simulated: Ditto interlocks carry `simulation_only: true` and `equipment_io: false`.

EVIDENCE

Claims are tested, run live, and checked again at runtime

211

backend tests (pytest) · Ruff · strict mypy

41

frontend tests (Vitest) · TypeScript · ESLint · build

5 + 5

routes at desktop and 390 px mobile, no browser errors

4 / 4

scenarios verified through the Ditto readback

Live scenario gate · scenario-smoke.sh

S1 reached Z04 EMERGENCY (1.52 % CH_4, tick 83) with the Ditto interlock verified; S3 read back E03 at 3.94 m^3/s with Z04 WARNING; S4 read back Z04 UNKNOWN_DATA; reset returned Ditto to tick 0.

Browser evidence · e2e/*.mjs

Headless Chromium on software WebGL (SwiftShader) with the lite profile; screenshots and Ditto JSON for every S1 stage and for S2, S3 and S4 in `mineguard/docs/final-*`.

Runtime validation · /evidence

Ditto-vs-model unit consistency, gas-balance residual bound, scenario repeatability, readback freshness and single-publication traceability, checked by the running service.

ON THE STAND

Ten clicks to see everything

1. Run `bash mineguard/scripts/bootstrap.sh`, open `http://127.0.0.1:8091`
2. Click **Z04 — Production face** in the zone list: CH_4, CO and airflow meters
3. Tray **S1 Gas Release**: ADVISORY → WARNING; the popup names the production stop
4. EMERGENCY: Safety Response shows MAIN and AUX **PENDING**, then **ACHIEVED 100 %**
5. **Reset**: Ditto returns to the tick-0 baseline
6. **S2 Fan Failure**: MAIN FAULTED, all airways 0 m^3/s → Reset
7. **S3 Airway Block**: network map draws E03 dashed amber “×250” → Reset
8. **S4 Sensor Loss**: Z04 turns purple, UNKNOWN_DATA → Reset
9. `/energy` → **Apply recommendation (simulated)**; fans read PENDING, then AT SETPOINT
10. `/twins` for the 16 Ditto twins · `/evidence` for live checks · Grafana `:3000` for history

LIMITS AND CLOSING

An honest model, built to answer “what happens if…”

Limits

Fictional, uncalibrated mine. Steady incompressible network flow without leakage, natural ventilation or reverse flow; well-mixed zones; demonstration thresholds; fan energy from the affinity law without motor or drive efficiency. 3D renders at a few frames per second on CPU-only WebGL — numbers, labels, the network map and controls do not depend on the frame rate.

Before any real use

Surveyed geometry, calibrated sensors and fans, an OT gateway, authentication and roles, redundancy, alarm management, a hazard analysis and regulatory approval. All actions here are simulated; nothing connects to equipment.

Methane is invisible. A connected twin makes its consequences visible — and keeps safety ahead of energy.

```
sudo apt-get update && sudo apt-get install -y make  ·  bash mineguard/scripts/bootstrap.sh  →  http://127.0.0.1:8091
```